

1:

```
> db.students.find({
  dept: "CS",
  marks: { $gt: 80 }
})
< [
  {
    _id: 1,
    name: 'Ananya',
    age: 21,
    dept: 'CS',
    marks: 88,
    grades: [
      'A',
      'B',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600001'
    },
    active: true
  },
  {
    _id: 3,
    name: 'Meena',
    age: 20,
    dept: 'CS',
    marks: 95,
    grades: [
      'A',
      'A',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600002'
    },
    active: true
  }
]
```

2:

```
> db.students.find({
  marks: {
    $gte: 60,
    $lte: 85
  }
})
< [
  {
    _id: 2,
    name: 'Arjun',
    age: 23,
    dept: 'IT',
    marks: 74,
    grades: [
      'B',
      'C',
      'B'
    ],
    address: {
      city: 'Madurai',
      pin: '625001'
    },
    active: true
  }
  {
    _id: 4,
    name: 'Ravi',
    age: 22,
    dept: 'ECE',
    marks: 61,
    grades: [
      'C',
      'C',
      'B'
    ],
  },
]
```

```
    address: {
      city: 'Trichy',
      pin: '620001'
    },
    active: false
  }
  {
    _id: 5,
    name: 'Kavya',
    age: 24,
    dept: 'IT',
    marks: 82,
    grades: [
      'A',
      'B',
      'B'
    ],
    address: {
      city: 'Coimbatore',
      pin: '641001'
    },
    active: true
  }
  {
    _id: 7,
    name: 'Priya',
    age: 22,
    dept: 'ECE',
    marks: 79,
    grades: [
      'B',
      'B',
      'A'
    ],
  },
]
```

```
    address: {
      city: 'Coimbatore',
      pin: '641001'
    },
    active: true
  }
  {
    _id: 7,
    name: 'Priya',
    age: 22,
    dept: 'ECE',
    marks: 79,
    grades: [
      'B',
      'B',
      'A'
    ],
    address: {
      city: 'Chennai',
      pin: '600003'
    },
    active: true
  }
  {
    _id: 9,
    name: 'Rakha',
    age: 21,
    dept: 'CS',
    marks: 68,
    grades: [
      'B',
      'C',
      'B'
    ],
    address: {
      city: 'Trichy',
      pin: '620002'
    },
  },
]
```

3:

```
> db.students.find({
  dept: { $in: ["IT", "ECE"] }
})
< [
  {
    _id: 2,
    name: 'Arjun',
    age: 23,
    dept: 'IT',
    marks: 74,
    grades: [
      'B',
      'C',
      'B'
    ],
    address: {
      city: 'Madurai',
      pin: '625001'
    },
    active: true
  },
  {
    _id: 4,
    name: 'Ravi',
    age: 22,
    dept: 'ECE',
    marks: 61,
    grades: [
      'C',
      'C',
      'B'
    ],
    address: {
      city: 'Trichy',
      pin: '620001'
    },
    active: true
  },
  {
    _id: 8,
    name: 'Suresh',
    age: 25,
    dept: 'IT',
    marks: 90,
    grades: [
      'A',
      'A',
      'B'
    ],
    address: {
      city: 'Madurai',
      pin: '625002'
    },
    active: false
  },
  {
    _id: 10,
    name: 'Vijay',
    age: 23,
    dept: 'ECE',
    marks: 45,
    grades: [
      'D',
      'F',
      'D'
    ],
    address: {
      city: 'Coimbatore',
      pin: '641002'
    },
    active: false
  }
]
```

4:

```
db.students.find(
  {
    dept: { $nin: ["CS"] }
  },
  {
    _id: 0,
    name: 1,
    dept: 1
  }
)
{
  name: 'Arjun',
  dept: 'IT'
}
{
  name: 'Ravi',
  dept: 'ECE'
}
{
  name: 'Kavya',
  dept: 'IT'
}
{
  name: 'Priya',
  dept: 'ECE'
}
{
  name: 'Suresh',
  dept: 'IT'
}
{
  name: 'Vijay',
  dept: 'ECE'
}
```

5:

```
> db.students.find(
  {
    dept: { $nin: ["CS"] }
  },
  {
    _id: 0,
    name: 1,
    dept: 1
  }
)
< {
  name: 'Arjun',
  dept: 'IT'
}
{
  name: 'Ravi',
  dept: 'ECE'
}
{
  name: 'Kavya',
  dept: 'IT'
}
{
  name: 'Priya',
  dept: 'ECE'
}
{
  name: 'Suresh',
  dept: 'IT'
}
{
  name: 'Vijay',
  dept: 'ECE'
}
```

```

> db.products.aggregate([
  {
    $group: {
      _id: "$category",
      totalRevenue: {
        $sum: {
          $multiply: ["$price", "$sold"]
        }
      }
    }
  }
])
< {
  _id: 'Books',
  totalRevenue: 629000
}
{
  _id: 'Electronics',
  totalRevenue: 16817000
}
{
  _id: 'Furniture',
  totalRevenue: 612000
}

```

6:

```

> db.students.aggregate([
  {
    $group: {
      _id: "$dept",
      totalStudents: { $sum: 1 },
      averageMarks: { $avg: "$marks" }
    }
  }
])
< {
  _id: 'ECE',
  totalStudents: 3,
  averageMarks: 61.666666666666664
}
{
  _id: 'CS',
  totalStudents: 4,
  averageMarks: 76.5
}
{
  _id: 'IT',
  totalStudents: 3,
  averageMarks: 82
}

```

7:

```
> db.students.updateMany(
  {},
  [
    $unset: { scholarship: "" }
  ]
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 10,
  modifiedCount: 0,
  upsertedCount: 0
}
```

8:

```
> db.students.updateMany(
  { dept: "CS" },
  [
    $set: { scholarship: false }
  ]
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
```

9:

```
> db.students.updateOne(
  { _id: 6 },
  [
    $inc: { marks: 10 }
  ]
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

10:

```

> db.students.find()
.sort({ marks: -1 })
.skip(2)
.limit(2)
< {
  _id: 1,
  name: 'Ananya',
  age: 21,
  dept: 'CS',
  marks: 88,
  grades: [
    'A',
    'B',
    'A'
  ],
  address: {
    city: 'Chennai',
    pin: '600001'
  },
  active: true,
  scholarship: false
}
{
  _id: 5,
  name: 'Kavya',
  age: 24,
  dept: 'IT',
  marks: 82,
  grades: [
    'A',
    'B',
    'B'
  ],
  address: {
    city: 'Coimbatore',
    pin: '641001'
  },
  active: true
}

```

11: }


```

db.employees.find({
  salary: { $gt: 50000 },
  experience: { $gt: 5 }
})
[
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd7e'),
    empId: 103,
    name: 'Rahul',
    age: 32,
    gender: 'Male',
    department: 'IT',
    salary: 75000,
    city: 'Mumbai',
    skills: [
      'Python',
      'MongoDB',
      'AWS'
    ],
    experience: 8,
    active: true
  }
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd80'),
    empId: 105,
    name: 'Karthik',
    age: 30,
    gender: 'Male',
    department: 'IT',
    salary: 68000,
    city: 'Chennai',
    skills: [
      'NodeJS',
      'MongoDB',
      'React'
    ],
    experience: 7,
    active: true
  }
]

```

12: }

```

> db.employees.find({
  $expr: {
    $gt: [
      { $size: "$skills" },
      2
    ]
  }
})
< [
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd7c'),
    empId: 101,
    name: 'Arun',
    age: 25,
    gender: 'Male',
    department: 'IT',
    salary: 45000,
    city: 'Chennai',
    skills: [
      'Java',
      'MongoDB',
      'Spring'
    ],
    experience: 3,
    active: true
  },
  {
    _id: ObjectId('6a26bce9015f1fd60bbfbd7e'),
    empId: 103,
    name: 'Rahul',
    age: 32,
    gender: 'Male',
    department: 'IT',
    salary: 75000,
    city: 'Mumbai',
    skills: [
      'Python',
      'MongoDB',
      'AWS'
    ],
    experience: 5,
    active: true
  }
]

```

13:

```

> db.employees.aggregate([
  {
    $group: {
      _id: "$department",
      avgSalary: { $avg: "$salary" }
    }
  },
  {
    $match: {
      avgSalary: { $gt: 50000 }
    }
  }
])
< {
  _id: 'IT',
  avgSalary: 62666.666666666664
}
{
  _id: 'HR',
  avgSalary: 55000
}

```

14:

```

> db.employees.aggregate([
  {
    $group: {
      _id: "$department",
      minSalary: { $min: "$salary" }
    }
  }
])
< {
  _id: 'HR',
  minSalary: 55000
}
{
  _id: 'IT',
  minSalary: 45000
}
{
  _id: 'Finance',
  minSalary: 50000
}
{
  _id: 'Marketing',
  minSalary: 42000
}

```

15:

```
> db.employees.updateMany(  
  {},  
  {  
    $rename: { city: "location" }  
  }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 6,  
  modifiedCount: 6,  
  upsertedCount: 0  
}
```

16:

```

> db.products.find({
  stock: { $type: "number" }
})
< [
  {
    _id: 101,
    name: 'Laptop',
    category: 'Electronics',
    price: 55000,
    stock: 20,
    tags: [
      'portable',
      'powerful'
    ],
    rating: 4.5,
    sold: 150
  },
  {
    _id: 102,
    name: 'Phone',
    category: 'Electronics',
    price: 18000,
    stock: 50,
    tags: [
      'portable',
      'compact'
    ],
    rating: 4.2,
    sold: 320
  },
  {
    _id: 103,
    name: 'Headphones',
    category: 'Electronics',
    price: 3500,
    stock: 80,
    tags: [
      'audio',
      'wireless'
    ],
    rating: 4,
    _id: 104,
    name: 'Desk Chair',
    category: 'Furniture',
    price: 8000,
    stock: 15,
    tags: [
      'ergonomic',
      'office'
    ],
    rating: 4.3,
    sold: 45
  },
  {
    _id: 105,
    name: 'Bookshelf',
    category: 'Furniture',
    price: 4200,
    stock: 30,
    tags: [
      'storage',
      'wood'
    ],
    rating: 3.8,
    sold: 60
  },
  {
    _id: 106,
    name: 'Python Book',
    category: 'Books',
    price: 650,
    stock: 100,
    tags: [
      'programming',
      'education'
    ],
    rating: 4.7,
    sold: 500
  },
  {

```

```
> db.students.updateMany(  
  { active: false },  
  {  
    $set: { active: true }  
  }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 3,  
  modifiedCount: 3,  
  upsertedCount: 0  
}
```

18: